

**Q1.**

The discrete random variable  $R$  has the following probability distribution.

|            |     |     |     |     |
|------------|-----|-----|-----|-----|
| $r$        | -2  | 0   | $a$ | 4   |
| $P(R = r)$ | 0.3 | $b$ | $c$ | 0.1 |

It is known that  $E(R) = 0.2$  and  $\text{Var}(R) = 3.56$

Find the values of  $a$ ,  $b$  and  $c$ .

(Total 4 marks)

**Q2.**

Aiden takes his car to a garage for its MOT test. The probability that his car will need to have  $X$  tyres replaced is shown in the table.

|            |     |      |      |     |     |
|------------|-----|------|------|-----|-----|
| $x$        | 0   | 1    | 2    | 3   | 4   |
| $P(X = x)$ | 0.1 | 0.35 | 0.25 | 0.2 | 0.1 |

- (a) Show that the mean of  $X$  is 1.85 and calculate the variance of  $X$ .

(4)

- (b) The charge for the MOT test is  $\pounds c$  and the cost of **each** new tyre is  $\pounds n$ . The total amount that Aiden must pay the garage is  $\pounds T$ .

- (i) Express  $T$  in terms of  $c$ ,  $n$  and  $X$ .

(1)

- (ii) Hence, using your results from part (a), find expressions for  $E(T)$  and  $\text{Var}(T)$ .

(4)

(Total 9 marks)

**Q3.**

In a computer game, players try to collect five treasures. The number of treasures that Isaac collects in one play of the game is represented by the discrete random variable  $X$ .

The probability distribution of  $X$  is defined by

$$P(X = x) = \begin{cases} \frac{1}{x+2} & x = 1, 2, 3, 4 \\ k & x = 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) (i) Show that  $k = \frac{1}{20}$ .

(2)

(ii) Calculate the value of  $E(X)$ .

(2)

(iii) Show that  $\text{Var}(X) = 1.5275$ .

(3)

(iv) Find the probability that Isaac collects more than 2 treasures.

(2)

(b) The number of points that Isaac scores for collecting treasures is  $Y$  where

$$Y = 100X - 50$$

Calculate the mean and the standard deviation of  $Y$ .

(4)

(Total 13 marks)

**Q4.**

(a) Joshua plays a game in which he repeatedly tosses an unbiased coin. His game concludes when he obtains either a head or 5 tails in succession.

The random variable  $N$  denotes the number of tosses of his coin required to conclude a game.

By completing **Table 1** below, calculate  $E(N)$ .

**Table 1**

| $n$        | 1 | 2 | 3             | 4 | 5              |
|------------|---|---|---------------|---|----------------|
| $P(N = n)$ |   |   | $\frac{1}{8}$ |   | $\frac{1}{16}$ |

(4)

(b) Joshua's sister, Ruth, plays a separate game in which she repeatedly tosses a coin that is **biased** in such a way that the probability of a head in a single toss of her coin is  $\frac{1}{4}$ . Her game also concludes when she obtains either a head or 5 tails in succession.

The random variable  $M$  denotes the number of tosses of her coin required to conclude her game.

Complete **Table 2** below.

**Table 2**

| $m$ | 1 | 2 | 3 | 4 | 5 |
|-----|---|---|---|---|---|
|-----|---|---|---|---|---|



|            |               |                |  |  |  |
|------------|---------------|----------------|--|--|--|
| $P(M = m)$ | $\frac{1}{4}$ | $\frac{3}{16}$ |  |  |  |
|------------|---------------|----------------|--|--|--|

(3)

- (c) (i) Joshua and Ruth play their games simultaneously. Calculate the probability that Joshua and Ruth will conclude their games in an equal number of tosses of their coins.

(5)

- (ii) Joshua and Ruth play their games simultaneously on 3 occasions. Calculate the probability that, on at least 2 of these occasions, their games will be concluded in an equal number of tosses of their coins. Give your answer to three decimal places.

(4)

(Total 16 marks)

**Q5.**

The random variable  $X$  has a binomial distribution with parameters  $n$  and  $p$ .

The mode,  $m$ , of  $X$  is such that

$$P(X = m) \geq P(X = m - 1) \quad \text{and} \quad P(X = m) \geq P(X = m + 1)$$

- (a) Use the **first** inequality to show that

$$m \leq (n + 1)p$$

(4)

- (b) Given that the **second** inequality results in

$$m \geq (n + 1)p - 1$$

deduce that the distribution  $B(10, 0.65)$  has one mode, and find the two values for the mode of the distribution  $B(35, 0.5)$ .

(3)

(Total 7 marks)

**Q6.**

The amount charged,  $\pounds X$ , for entry to a museum depends on the status of the visitor. The following table shows the charges together with the probability that a visitor will have a particular status.

| Status         | Charge, $\pounds x$ | $P(X = x)$ |
|----------------|---------------------|------------|
| Child under 16 | 1.00                | 0.35       |
| Student        | 1.50                | 0.21       |
| Senior citizen | 2.00                | 0.24       |

|       |      |      |
|-------|------|------|
| Adult | 3.00 | 0.20 |
|-------|------|------|

- (a) For entrance charges paid by visitors to the museum calculate:
- the mean;
  - $E(X^2)$ ;
  - the standard deviation.
- (6)
- (b) Find the probability that the charge for a randomly selected visitor will be greater than or equal to:
- the mean;
  - the mode.
- (4)
- (c) Children under 5 are admitted free and have been omitted from the probability distribution shown above. If they were included in the probability distribution, explain whether:
- the mean would increase, stay the same or decrease;
  - the standard deviation would increase, stay the same or decrease.
- (4)
- (Total 14 marks)

### Q7.

A village inn offers bed and breakfast and has four bedrooms available for customers. The number of bedrooms occupied each night may be modelled by the random variable,  $X$ , with the following probability distribution.

| $x$ | $P(X = x)$ |
|-----|------------|
| 0   | 0.22       |
| 1   | 0.25       |
| 2   | 0.20       |
| 3   | 0.14       |
| 4   | 0.19       |

- (a) Calculate:
- the mean of  $X$ ;
  - $E(X^2)$ ;
  - the standard deviation of  $X$ .

- (b) Find the probability that the number of occupied bedrooms:
- (i) is less than 3;
  - (ii) exceeds the mean;
  - (iii) exceeds the median.

(5)

(Total 11 marks)



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